

Patrick Selby

psselby@gsumail.gram.edu • (614) 332-6860 • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

EDUCATION

Grambling State University

Grambling, LA

B.S. Cybersecurity — Minor: CIS | GPA: 3.83/4.0 | Jan 2025 – Dec 2028

Coursework: Attacks, Threats & Vulnerabilities (Sec+ SYO-701) • Discrete Structures • Data Structures & Algorithms • Probability & Statistics • Calculus I

EXPERIENCE

Research Assistant — Mobile Security & Applied Computing

Mar 2026 – May 2026

Grambling State University • AIoT Lab, Dr. Vasanth Iyer • Grambling, LA

- **Reduced QR phishing exposure** to zero cloud dependency by building ScanSafe, an on-device Android app using an **18-rule heuristic URL analysis engine** with no pretrained models
- **Improved phishing detection coverage by 50%** by expanding URL risk engine from 12 to 18 rules; Rule 13 directly motivated by a live GSU-targeted attack impersonating Microsoft Office 365 via SafeLinks wrapping

Utilized: Android Studio, Python, Git, GitHub

IoT Cyber Defense Extern

Feb 2026 – Mar 2026

Hydroficient (via Extern) • Remote • [Certificate of Completion](#)

- **Built end-to-end IoT security pipeline** covering STRIDE threat modeling, TLS/mTLS device authentication on MQTT, **3-layer replay defense** (timestamps, sequence counters, HMAC), and Isolation Forest anomaly detection

Utilized: Python, MQTT, OpenSSL, Wireshark, Linux CLI

Intermediate Cybersecurity Fellow

Feb 2026 – May 2026

CodePath • Remote • Certificate of Achievement with Honors • ID: 397157

- **Completed 7 hands-on security labs** covering Wireshark/SMTP forensics, Linux auditd monitoring, FTP directory traversal, DoS/Slowloris mitigation, **Splunk SIEM** analysis, CSIRT incident response, and SolarWinds IOC threat hunting
- **Led capstone team** through full incident response lifecycle — dataset analysis, playbook selection, IOC identification, and **live Demo Day presentation** to a technical audience

Utilized: Wireshark, auditd, Splunk, nginx, VirusTotal, Catalyst, Linux CLI, Python

PROJECTS

Linux Security Monitoring & Incident Response

[View Project](#)

auditd • ausearch • Linux CLI • Log Analysis

- **Delivered root cause analysis with 3 actionable control recommendations** by configuring auditd watch rules, tracing file modification events to specific processes via ausearch, and documenting chain-of-custody evidence

IT Risk Assessment & Data Classification

[View Project](#)

Risk Analysis • Asset Inventory • Data Classification • Security Controls

- **Prioritized 5+ security controls** across a simulated multi-department organization by inventorying assets, classifying data sensitivity tiers, and conducting likelihood-impact analysis

ScanSafe — On-Device QR Phishing Detection

[View Project](#)

Android • Heuristic URL Analysis • Python • Git

- **Eliminated cloud dependency and false-negative risk** by engineering an on-device **18-rule** heuristic URL analysis engine that detects QR phishing in real time without pretrained models

TECHNICAL SKILLS

Cybersecurity	Linux CLI, Nmap, Wireshark, Vulnerability Assessment, Threat Modeling, Log Analysis
Protocols	MQTT, TLS/mTLS, HMAC, Public/Private Key Cryptography, TCP/IP
Forensics	auditd / ausearch, Network Forensics, Log Correlation, Splunk
Programming	Python, SQL Git, GitHub
Cloud	IoT Device Security, AWS Cloud Foundations
Research	STRIDE, Risk Assessment, STEEP Analysis, Technical Writing

CERTIFICATIONS

CompTIA Security+ (In Progress — SYO-701) • [AWS Academy Cloud Foundations](#) • CodePath Intermediate Cybersecurity (CYB102) — **Certificate of Achievement with Honors**

[Google: Foundations of Cybersecurity](#) • [Play It Safe: Manage Security Risks](#)

IBM SkillsBuild: [Getting Started with AI](#) • [Getting Started with Cybersecurity](#) • [Getting Started with Data](#) • [Working in a Digital World](#)

AFFILIATIONS & PROFESSIONAL ENGAGEMENT

Selected Participant — 2026 IOBSE Conference (Apr 14–16) • International Organization of Black Security Executives

CodePath • ColorStack • NSBE • IEEE • Honor Society® • S.E.C.U.R.E. Cybersecurity Club (GSU) • NSLS • Cloud Security Alliance • ASIS International